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Thesis Validation Letter

Validation letter

Acknowledgements

bismi al-ll_ahi al-rra.hm_ani al-rra.hImi

Alhamdullahirabbil'alamin, finally got to this point, a splash of success that You gave me Rabb Allah SWT is the Almighty, endlessly thanking you. Prayers and greetings to the Prophet Muhammad SAW and his noble friends.

In accordance with the words of the Prophet Muhammad SAW that when we have an advantage we might help the other people, as mentioned in hadith narrated by Ath-Thabrani:

a.habba Al-nnaasi 'il_A al-ll_ahi ta'aa l_A aanfa'uhum lil-nnaasi..

"The most loved Man/Woman by Allah is the most beneficial to humans beings.." (Narrated by Ath-Thabrani in Al Mu'jam Al Kabir no. 13280).

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摘要

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Abstract

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Keywords: keyword1, keyword2, keyword3.



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Chapter 1

Introduction

1.1 Background

Wireless communication systems continue to require higher data rates, wider coverage, and more reliable link quality. These requirements become more challenging as communication systems operate at higher frequencies, where propagation loss, blockage, antenna alignment, and coverage limitation can strongly affect the received signal. In this condition, the antenna is not only a radiating component, but also an important part of the overall communication channel because its gain, radiation pattern, polarization, and beam direction directly influence the received power and link performance.

Directional antennas are widely used to improve link quality by concentrating electromagnetic energy toward a desired direction. However, conventional high-gain antenna structures can be bulky, complex, or difficult to integrate into compact communication platforms. A metasurface transmitarray lens antenna offers an attractive alternative because it can provide spatial phase control using a planar arrangement of engineered unit cells. By controlling the transmission phase across the aperture, the transmitarray lens can transform the incident wave from a feed antenna into a focused or collimated wavefront, which can improve gain and support directional communication.

The performance of a metasurface transmitarray lens antenna depends on both the electromagnetic behavior of its unit cells and the system-level effect of the resulting radiation pattern. Therefore, antenna-level evaluation alone is not sufficient to understand its usefulness in a communication scenario. Parameters such as transmission magnitude, phase response, gain, directivity, beam direction, and side-lobe level must be connected with channel-related metrics such as received power, path loss, signal-to-noise ratio, and coverage. This relationship is important because an antenna design that performs well in isolation may still produce different link behavior depending on propagation distance, antenna orientation, and environmental assumptions.

Based on this background, this thesis investigates a metasurface transmitarray lens antenna through simulation and communication channel evaluation. ANSYS HFSS is used to evaluate the electromagnetic characteristics of the antenna, while channel-level analysis is used to observe how the antenna performance affects the communication link. The study focuses on building a simulation workflow that connects the physical antenna design with practical communication performance indicators.

1.2 Motivation

The main motivation of this research is the need to evaluate antenna designs from both electromagnetic and communication perspectives. In many antenna studies, the evaluation is mainly focused on parameters such as reflection coefficient, gain, directivity, and radiation pattern. Although these parameters are essential,

they do not fully describe how the antenna affects the received signal in an actual communication link. For directional antennas such as metasurface transmitarray lenses, this gap becomes important because beam focusing, phase compensation, and aperture efficiency can directly change the channel response.

Another motivation is the practical need for compact and efficient high-gain antenna structures. A metasurface transmitarray lens antenna can provide beam-forming capability with a relatively low-profile planar structure. This makes it a promising candidate for communication systems that require directional radiation while maintaining simpler integration compared with bulky lens or reflector structures. However, the design must be carefully evaluated because unit cell loss, limited phase coverage, feed illumination, and phase error can reduce the final antenna performance.

This thesis is also motivated by the need for a clear simulation procedure that links unit cell design, full antenna simulation, and communication channel evaluation. By combining these stages, the antenna can be assessed not only by its radiation characteristics but also by its influence on link-related metrics such as received power, path loss, signal-to-noise ratio, and coverage. This approach provides a more complete understanding of whether the designed metasurface transmitarray lens antenna can support the intended communication scenario.

1.3 Thesis Structure

This thesis is organized as follows. Chapter 1 introduces the research background, motivation, and thesis structure. Chapter 2 presents the theoretical

background of metasurface transmitarray lens antennas and communication channel evaluation. Chapter 3 describes the simulation setup and design workflow, including unit cell design, transmitarray lens configuration, ANSYS HFSS simulation, and channel evaluation setup. Chapter 4 presents the simulation results and discussion for both antenna-level and communication-level performance. Chapter 5 summarizes the conclusions, main contributions, limitations, and future work.

Chapter 2

Theoretical Background

2.1 Antenna Theory

2.2 Metasurface Transmitarray Lens Antenna

This section presents a paraphrased synthesis of the transmitarray antenna concepts discussed by Abdelrahman et al. [1] and relates them to metasurface transmitarray lens antennas. A transmitarray antenna consists of an illuminating feed source and a thin transmitting surface composed of antenna elements or metasurface unit cells. The feed is placed at an equivalent focal point, causing the incident wave on the aperture to have a spherical phase front.

Each element on the transmitting surface is designed with a specific transmission coefficient to control the transmitted phase. By compensating for the different path lengths from the feed to each element, the transmitarray converts the spherical phase front into a more planar transmitted phase front and forms a directive beam with higher gain. This makes transmitarray antennas useful for compact wireless communication, remote sensing, terahertz imaging, and other applications requiring controlled high-gain radiation.

2.2.1 Unit-Cell Element Analysis

The performance of a transmitarray is strongly influenced by the design of its individual unit-cell elements, since each element must provide a controlled transmission phase while maintaining adequate transmission magnitude. In a meta-surface transmitarray lens antenna, the unit cell is therefore analyzed not only as a physical geometry, but also as a phase-control element that determines how the incident wave from the feed is transformed after passing through the aperture.

The transmission characteristics of a unit cell are commonly evaluated through a parametric study, where selected geometrical dimensions are varied and the resulting phase and magnitude responses are observed. In electromagnetic simulation, the unit cell can be modeled with periodic boundary conditions to approximate its behavior inside a larger array. This approach allows the designer to estimate the phase range, transmission loss, and sensitivity of the element before constructing the complete transmitarray aperture. A suitable unit-cell design should provide a broad phase variation, preferably close to a full 360° range, while keeping the transmission magnitude as high and stable as possible [1].

In this thesis, the unit-cell candidate is selected using the power conservation principle, where the incident power is divided into reflection, transmission, and loss:

$$|S_{11}|^2 + |S_{21}|^2 + P_{\text{loss}} = 1, \quad (2.1)$$

$$P_R = |S_{11}|^2 = 10^{S_{11,\text{dB}}/10}, \quad P_T = |S_{21}|^2 = 10^{S_{21,\text{dB}}/10}. \quad (2.2)$$

Here, S_{11} and S_{21} represent the reflection and transmission coefficients, respectively.

Therefore, the preferred element is the one with high transmitted power, low reflected power, low loss, and the required transmission phase response.

2.2.2 Antenna Lens Design

The antenna lens design in this thesis is formulated based on phase compensation principles and transmitarray lens design approaches reported in [2, 3, 1]. The main purpose of the lens is to compensate for the phase difference of the spherical wavefront radiated by the feed source. Because the feed is placed at a finite focal distance, the incident wave reaches different positions on the aperture through different path lengths. Therefore, the aperture must provide a spatially varying transmission phase so that the transmitted fields become approximately phase-aligned after passing through the lens. This design concept is illustrated in Fig. 2-1.

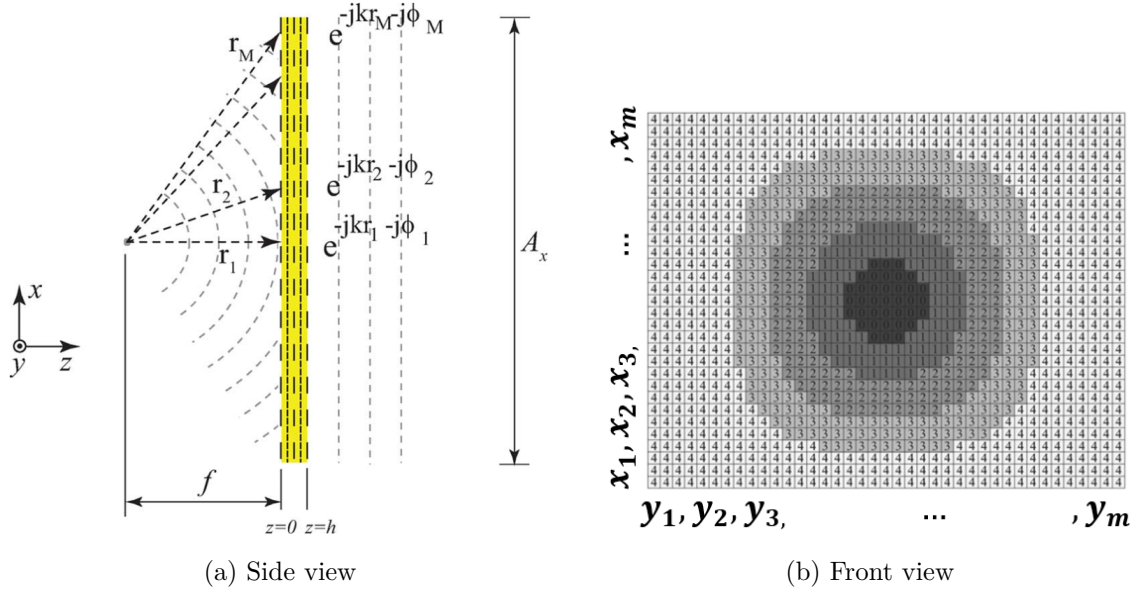


Figure 2-1: Antenna lens illustrations showing the feed-to-aperture path-length differences in the side view and the aperture zoning in the front view.

Figure 2-1(a) shows the side view of the lens configuration, where the feed illuminates the aperture from the focal position and produces different propagation paths toward different parts of the lens. Figure 2-1(b) shows the front view of the

aperture, where the lens is implemented as a two-dimensional unit-cell array. The numbered regions indicate that the aperture is grouped into concentric phase zones, so unit cells within the same zone can use the same phase-shifter type. This discretized zoning approximates the continuous phase profile required for lens operation while keeping the physical implementation simpler.

For the phase calculation, the aperture is divided into M concentric zones indexed by m , where $m = 0, 1, \dots, M - 1$. If the center of zone m is located at $(x_m, y_m, z = 0)$, the required transmission phase for the unit cells in that zone is calculated as

$$\Phi(x_m, y_m) = -k_0(r_m - f) + k_0 \left(\sqrt{(A/2)^2 + f^2} - f \right) + \Phi_0. \quad (2.3)$$

In this expression, $\Phi(x_m, y_m)$ denotes the desired transmission phase at the center of zone m , $k_0 = 2\pi/\lambda_0$ is the free-space wavenumber, λ_0 is the free-space wavelength, r_m is the distance from the feed point to the center of zone m , f is the focal distance, A is the aperture size, and Φ_0 is a positive constant phase delay applied to all unit-cell responses. The resulting phase values are then used as the design targets for selecting and arranging the unit-cell elements across the lens aperture.

2.3 Wireless Channel

This thesis evaluates the wireless channel to examine how the antenna lens modifies the end-to-end electromagnetic response between multiple transmit and receive positions. Following the general wireless communication framework described by Tse and Viswanath [4], the analysis is not limited to received power, but also con-

siders how the lens affects angular separation, spatial multiplexing behavior, and the correlation between propagation paths. For this purpose, the measured or simulated S-parameters are treated as a frequency-domain representation of the communication channel, allowing the with-lens and without-lens cases to be compared in a consistent manner.

2.3.1 Formulation Wireless Channel

In a general multiple-input multiple-output (MIMO) system, the wireless channel connects N_T transmit ports to N_R receive ports. At each frequency point f , the channel response can be represented by a complex matrix

$$\mathbf{H}(f) \in \mathbb{C}^{N_R \times N_T}, \quad (2.4)$$

where each row corresponds to a receive port and each column corresponds to a transmit port. When the channel is obtained from electromagnetic simulation or network-analyzer measurement, each element of the matrix can be constructed from the transmission S-parameter between the corresponding transmit and receive ports:

$$H_{ij}(f) = S(\text{rx}_i, \text{tx}_j; f), \quad (2.5)$$

where $H_{ij}(f)$ denotes the channel coefficient from the j -th transmit port to the i -th receive port at frequency f .

The corresponding frequency-domain input-output relationship is written

as

$$\mathbf{y}(f) = \mathbf{H}(f)\mathbf{x}(f) + \mathbf{n}(f), \quad (2.6)$$

where $\mathbf{x}(f) \in \mathbb{C}^{N_T \times 1}$ is the transmitted signal vector, $\mathbf{y}(f) \in \mathbb{C}^{N_R \times 1}$ is the received signal vector, and $\mathbf{n}(f)$ is the additive noise vector. For example, in a 3×3 configuration, the channel matrix becomes

$$\mathbf{H}(f) = \begin{bmatrix} S(\text{rx}_1, \text{tx}_1; f) & S(\text{rx}_1, \text{tx}_2; f) & S(\text{rx}_1, \text{tx}_3; f) \\ S(\text{rx}_2, \text{tx}_1; f) & S(\text{rx}_2, \text{tx}_2; f) & S(\text{rx}_2, \text{tx}_3; f) \\ S(\text{rx}_3, \text{tx}_1; f) & S(\text{rx}_3, \text{tx}_2; f) & S(\text{rx}_3, \text{tx}_3; f) \end{bmatrix}. \quad (2.7)$$

In this formulation, $\mathbf{H}(f)$ represents the measured or simulated end-to-end RF channel response. It therefore includes not only free-space propagation, but also the effects of the transmit antenna, receive antenna, antenna lens response, propagation path, mutual coupling, impedance mismatch, and measurement reference planes. This makes the S-parameter-based matrix suitable for empirical channel evaluation in the antenna-lens-assisted MIMO system.

2.3.2 Singular Value Decomposition of the Channel Matrix

After the channel matrix is constructed, singular value decomposition (SVD) is applied to $\mathbf{H}(f)$ at each frequency point. This decomposition expresses the MIMO channel as

$$\mathbf{H}(f) = \mathbf{U}(f)\mathbf{\Sigma}(f)\mathbf{V}^H(f), \quad (2.8)$$

where $\mathbf{U}(f)$ and $\mathbf{V}(f)$ are unitary matrices, $\mathbf{V}^H(f)$ denotes the Hermitian transpose of $\mathbf{V}(f)$, and $\mathbf{\Sigma}(f)$ contains the singular values of the channel matrix. For the 3×3

channel example used in this thesis, the singular-value matrix can be written as

$$\mathbf{\Sigma}(f) = \text{diag}(\sigma_1(f), \sigma_2(f), \sigma_3(f)), \quad (2.9)$$

with

$$\sigma_1(f) \geq \sigma_2(f) \geq \sigma_3(f) \geq 0. \quad (2.10)$$

The singular value $\sigma_i(f)$ represents the amplitude gain of the i -th equivalent parallel MIMO eigenchannel, while $\sigma_i^2(f)$ represents its corresponding power gain. If only $\sigma_1(f)$ is dominant, the channel is mainly supported by one spatial mode. In contrast, when $\sigma_1(f)$, $\sigma_2(f)$, and $\sigma_3(f)$ have comparable values, the channel provides better spatial multiplexing capability.

Two additional indicators are extracted from the same singular values. The condition number is used to evaluate the balance among the MIMO eigenchannels:

$$\kappa(f) = \frac{\sigma_1(f)}{\sigma_3(f)}. \quad (2.11)$$

A lower $\kappa(f)$ indicates that the channel modes are more balanced. The effective rank is then used to estimate the effective number of active spatial modes:

$$p_i(f) = \frac{\sigma_i^2(f)}{\sum_{k=1}^3 \sigma_k^2(f)}, \quad r_{\text{eff}}(f) = \exp \left(- \sum_{i=1}^3 p_i(f) \ln p_i(f) \right). \quad (2.12)$$

For a 3×3 channel, $r_{\text{eff}}(f)$ ranges from 1 to 3. A value closer to 3 indicates that more spatial modes contribute to the channel response, while a value closer to 1 indicates that the channel is dominated by fewer modes.

Chapter 3

Simulation Setup and Design

3.1 Overview

This chapter describes the simulation setup and design procedure used in this thesis. The workflow starts from the definition of the design requirements, followed by the metasurface unit cell design, transmitarray lens configuration, antenna simulation setup, and communication channel evaluation. The simulation is intended to evaluate both the electromagnetic performance of the antenna and its influence on the communication channel.

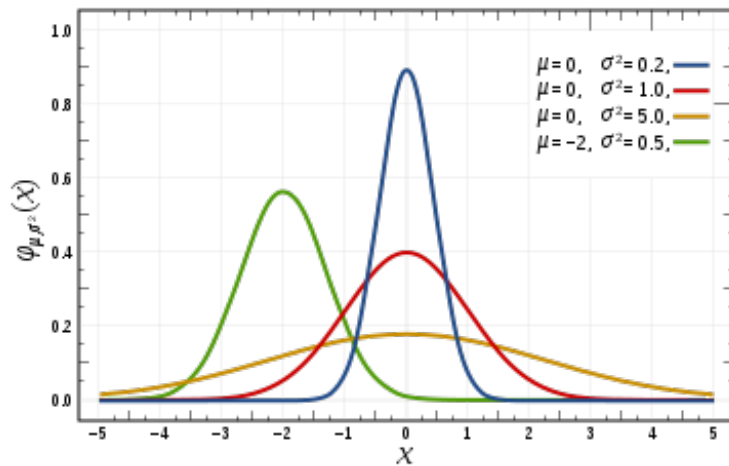


Figure 3-1: Example simulation-related figure for Chapter 3

3.2 Design Requirements

The design requirements define the target performance and simulation scope. The main parameters include operating frequency, aperture size, feed position, focal distance, polarization, desired beam direction, gain target, and channel evaluation scenario. These requirements are used as the basis for the transmitarray lens design and the communication link evaluation.

3.3 Metasurface Unit Cell Design

The metasurface unit cell is designed to provide the required transmission phase response while maintaining acceptable transmission magnitude. Each unit cell geometry is simulated to obtain its transmission coefficient, phase response, and operating bandwidth. The unit cell response is then used to determine the phase compensation required across the transmitarray aperture.

3.3.1 Unit Cell Boundary Condition

Periodic boundary conditions are used to represent an infinite metasurface array during unit cell simulation. The incident wave is applied to the unit cell, and the transmitted response is observed at the output port. This setup allows the phase and magnitude response of one unit cell to be evaluated efficiently.

3.3.2 Transmission Phase Response

The transmission phase response is the main design parameter for the transmitarray lens. A suitable unit cell should provide a wide phase tuning range

with low transmission loss. The selected unit cell dimensions are mapped to the required phase distribution of the aperture.

3.4 Transmitarray Lens Design

The transmitarray lens is formed by arranging unit cells over a planar aperture. Each unit cell is assigned a phase shift based on its position relative to the feed antenna. The purpose of the lens is to compensate for the feed path difference and produce a desired transmitted wavefront.

3.4.1 Feed Antenna Configuration

The feed antenna is placed at a selected focal distance from the transmitarray aperture. The feed position determines the incident phase distribution on the lens surface. Proper feed placement is important to achieve efficient aperture illumination and reduce spillover loss.

3.4.2 Phase Compensation Distribution

The phase compensation distribution is calculated from the path length between the feed antenna and each unit cell position. The required phase is then converted into a corresponding unit cell geometry. This process determines the final layout of the metasurface transmitarray lens.

3.5 ANSYS HFSS Simulation Setup

ANSYS HFSS is used to evaluate the electromagnetic performance of the antenna design. The simulation setup includes material definition, geometry construction, excitation, boundary conditions, radiation region, meshing configuration, and frequency sweep. The antenna performance is evaluated using parameters such as S-parameters, gain, directivity, radiation pattern, and aperture behavior.

3.5.1 Material and Geometry Definition

The substrate, conductor, and surrounding air region are defined based on the selected antenna structure. The unit cell and transmitarray aperture geometry are constructed according to the final design dimensions.

3.5.2 Excitation and Boundary Condition

The excitation is assigned according to the antenna feeding structure. Radiation or open boundary conditions are applied around the simulation domain to represent free-space propagation. For unit cell simulation, periodic boundaries are used, while for full antenna simulation, an open radiation region is applied.

3.5.3 Frequency Sweep and Mesh Setup

The frequency sweep is configured around the operating frequency to observe antenna performance across the target bandwidth. Mesh refinement is applied to ensure that the electromagnetic field distribution and radiation characteristics are accurately captured.

3.6 Communication Channel Evaluation Setup

The communication channel evaluation is used to observe how the antenna performance affects the communication link. The antenna radiation pattern, gain, and beam direction are used as input parameters for channel evaluation. The channel is assessed using link-related quantities such as received power, path loss, coverage, and signal-to-noise ratio.

3.6.1 Evaluation Scenario

The evaluation scenario defines the transmitter position, receiver position, propagation distance, antenna orientation, and surrounding environment. This scenario is used to compare the channel behavior under different antenna or propagation conditions.

3.6.2 Ray Tracing or Hybrid Evaluation

Ray tracing or a hybrid HFSS-based evaluation can be used to estimate the communication channel. In the hybrid approach, HFSS provides accurate antenna characteristics, while the propagation evaluation estimates how the radiated wave reaches the receiver. This allows antenna-level results to be connected with communication channel performance.

3.6.3 Channel Performance Metrics

The main channel performance metrics include path loss, received power, signal-to-noise ratio, delay spread, angular spread, and coverage area. These metrics

are used to evaluate whether the designed antenna improves the communication link in the selected scenario.

3.7 Simulation Workflow

The complete simulation workflow is summarized as follows. First, the design requirements are defined. Second, the unit cell is simulated and optimized. Third, the transmitarray lens phase distribution is calculated and implemented. Fourth, the full antenna is simulated in HFSS. Finally, the antenna results are used to evaluate the communication channel performance.

Chapter 4

Simulation Results and Discussion

4.1 Overview

This chapter presents the simulation results obtained from the metasurface transmitarray lens antenna design and the communication channel evaluation. The results are organized into antenna-level electromagnetic performance and communication-level channel performance. The purpose of this chapter is to evaluate whether the designed antenna can support the required communication link.

4.2 Unit Cell Simulation Results

The unit cell simulation results are used to verify the transmission magnitude and phase response of the selected metasurface element. The unit cell should provide sufficient phase coverage while maintaining low transmission loss across the target operating frequency.

4.2.1 Transmission Magnitude

The transmission magnitude result shows how much electromagnetic power passes through the unit cell. A high transmission magnitude is desired because it indicates lower insertion loss and better aperture efficiency.

4.2.2 Transmission Phase

The transmission phase result is used to map the required lens phase compensation to the corresponding unit cell geometry. The phase response should cover the required phase range with smooth variation between design states.

Table 4-1: Summary of unit cell simulation results

Parameter	Target/Condition	Simulation Result
Operating frequency	TBD	TBD
Transmission magnitude	TBD	TBD
Transmission phase range	TBD	TBD
Unit cell size	TBD	TBD
Bandwidth	TBD	TBD

4.3 Transmitarray Lens Simulation Results

The full transmitarray lens simulation evaluates the antenna radiation performance after the unit cells are arranged over the aperture. This analysis shows whether the lens can transform the feed radiation into the desired directional beam.

4.3.1 Reflection Coefficient

The reflection coefficient is evaluated to observe impedance matching near the operating frequency. A lower reflection coefficient indicates that more input power is accepted by the antenna system.

4.3.2 Radiation Pattern

The radiation pattern shows the angular distribution of radiated power. The main beam direction, beamwidth, and side-lobe level are observed to evaluate

the focusing and beam-forming performance of the transmitarray lens.

4.3.3 Gain and Directivity

Gain and directivity are used to evaluate the ability of the transmitarray lens antenna to concentrate electromagnetic energy in the desired direction. A higher gain indicates better suitability for directional communication links.

4.3.4 Aperture Efficiency

Aperture efficiency describes how effectively the physical aperture is converted into radiated gain. This parameter is influenced by transmission loss, phase error, feed illumination, spillover, and aperture taper.

Table 4-2: Summary of transmitarray antenna simulation results

Parameter	Simulation Result
Reflection coefficient	TBD
Peak gain	TBD
Directivity	TBD
Main beam direction	TBD
Half-power beamwidth	TBD
Side-lobe level	TBD
Aperture efficiency	TBD

4.4 Communication Channel Evaluation Results

The communication channel evaluation connects the antenna simulation results with link-level performance. The antenna radiation pattern and gain are used to evaluate how the designed antenna affects received power, path loss, coverage, and link quality.

4.4.1 Evaluation Scenario

The evaluation scenario includes the transmitter location, receiver location, antenna orientation, propagation distance, and environmental assumptions. These conditions are used to interpret the channel results consistently.

4.4.2 Received Power

Received power is used to evaluate how much signal power is available at the receiver. The result is affected by transmit power, antenna gain, propagation loss, antenna alignment, and multipath propagation.

4.4.3 Path Loss

Path loss describes the reduction of signal power between transmitter and receiver. The evaluated path loss is compared across the selected propagation condition to determine whether the antenna provides sufficient link margin.

4.4.4 Signal-to-Noise Ratio

Signal-to-noise ratio is used to estimate the reliability of the communication link. A higher signal-to-noise ratio indicates better received signal quality and improved communication performance.

4.4.5 Coverage Evaluation

Coverage evaluation is used to observe the spatial region where the communication link can satisfy the required performance. The coverage result can be

used to compare antenna configurations, beam directions, or propagation scenarios.

Table 4-3: Summary of communication channel evaluation results

Parameter	Evaluation Result
Propagation distance	TBD
Received power	TBD
Path loss	TBD
Signal-to-noise ratio	TBD
Delay spread	TBD
Coverage area	TBD

4.5 Comparison and Discussion

The simulation results are discussed by comparing antenna performance and channel performance. The relationship between antenna gain, radiation direction, path loss, and received power is analyzed to understand the impact of the metasurface transmitarray lens antenna on the communication channel.

4.6 Summary

This chapter summarizes the main simulation findings. The unit cell results verify the phase and transmission behavior of the metasurface element. The antenna results evaluate radiation performance, while the communication channel results show the link-level effect of the designed antenna.

Chapter 5

Conclusion

5.1 Summary

This thesis investigates the use of a metasurface transmitarray lens antenna for communication channel evaluation. The theoretical background covers the metasurface transmitarray concept, antenna performance parameters, and communication channel evaluation. The simulation setup describes the unit cell design, transmitarray lens configuration, ANSYS HFSS simulation, and channel evaluation workflow.

5.2 Conclusion

The simulation and evaluation framework provides a way to connect antenna-level electromagnetic performance with communication-level channel behavior. The metasurface transmitarray lens antenna is evaluated through parameters such as transmission response, radiation pattern, gain, directivity, received power, path loss, and signal-to-noise ratio. These results are used to determine how the antenna design affects the communication link performance.

5.3 Main Contributions

The main contributions of this thesis can be summarized as follows:

- The theoretical basis of metasurface transmitarray lens antennas and communication channel evaluation is reviewed.
- A simulation workflow for the metasurface transmitarray lens antenna is defined using ANSYS HFSS.
- The antenna simulation results are connected with channel-level performance metrics.
- The communication channel is evaluated based on link-related parameters such as received power, path loss, signal-to-noise ratio, and coverage.

5.4 Limitations

The evaluation is limited by the assumptions used in the antenna simulation and communication channel scenario. Material properties, fabrication tolerance, antenna alignment, environmental simplification, and propagation assumptions may affect the final results. Therefore, further validation is required when measurement data or a more detailed propagation environment becomes available.

5.5 Future Work

Future work may include measurement validation of the fabricated transmitarray lens antenna, more detailed channel evaluation in realistic environments,

comparison between ray tracing and hybrid electromagnetic approaches, and optimization of the antenna design for improved link performance.

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